1. Find the general solution.

   (a) $xy' - 2y = -x$.

   (b) $y' = \frac{1+2y^2}{y \sin x}$.

   (c) $xy' - y = 2x \ln x$.

   (d) $xy' + 2y = \frac{\cos x}{x}$.

   (e) $y' + 2xy = xe^{-x^2}$.

   (f) $y' = xy^2 - x - y^2 + 1$.

   (g) $e^y \sin 2x dx + \cos x(e^{2y} - y)dy = 0$.

   (h) $y' = x\sqrt{\frac{1-y^2}{1-x^2}}$.

2. Find all functions that satisfy the differential equation $y' - y = y'' - y'$. Hint: Set $z = y' - y$.

3. Suppose that a certain population P has a birth rate dB/dt and a death rate dD/dt. Then the rate of change of P is the difference

$$\frac{dP}{dt} = \frac{dB}{dt} - \frac{dD}{dt}.$$

   (a) Assume that $dB/dt = aP$ and $dD/dt = bP$, where a and b are constants. Find $P(t)$ if $P(0) = P_0 > 0$.

   (b) What happens to $P(t)$ as $t \to \infty$ if

      i. $a > b$,   ii. $a = b$,   iii. $a < b$?

4. An advertising company introduces a new product to a metropolitan area of population M. Let $P = P(t)$ denote the number of people who become aware of the product by time t. Suppose that P increases at a rate which is proportional to the number of people still unaware of the product. The company determines that no one was aware of the product at the beginning of the campaign [$P(0) = 0$] and that 30% of the people were aware of the product after 10 days of advertising.

   (a) Give the differential equation that describes the number of people who become aware of the product by time t.

   (b) Determine the solution of the differential equation from part (a) that satisfies the initial condition $P(0) = 0$.

   (c) How long does it take for 90% of the population to become aware of the product?

5. A flu virus is spreading rapidly through a small town with a population of 25,000. The virus is spreading at a rate proportional to the product of the number of people who have been infected and the number who haven't been infected. When first reported, at time $t = 0$, 100 people had been infected and 10 days later, 400 people.

   (a) How many people will have been infected by time t? (Measure t in days.)

   (b) How long will it take for the infection to reach half of the population?

   (c) Use a graphing utility to graph the function you found in part (a).